

Harshini Murugadoss

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TECHNICAL SKILLS

AI & Orchestration: Agentic AI (LangGraph, LangChain), RAG, Vector DBs (OpenSearch, ChromaDB), Hybrid Search, LLM Evaluation (Langfuse), LLM Security/Red Teaming
Backend & Infrastructure: FastAPI, PostgreSQL (Supabase), Redis, Apache Airflow, Docker, AWS, GCP, NVIDIA Triton, CI/CD
Frontend & UI: React 18, State Management, Responsive UI, Supabase Auth
Languages: Python, TypeScript, SQL, HTML/CSS

EXPERIENCE

AI Operations Engineer

May 2026 – Jul 2026

Enkrypt AI, Hybrid

- Engineered adversarial testing frameworks (prompt injection, data leaks) to stress-test enterprise LLM systems, delivering technical vulnerability mappings that directly drove model hardening decisions.
- Developed interactive security simulation modules on Enkrypt AI's Academy, a self-learning platform covering LLM risk, prompt injection, and secure generative AI adoption for both technical and executive audiences.

Full-Stack AI Engineer

Jun 2025 – Apr 2026

India Literacy Project, Remote

- Owned and shipped a production full-stack platform end-to-end (React 18 + TypeScript, FastAPI, PostgreSQL (Supabase) database) serving **100+** concurrent K-12 students across **7** Indian states.
- Architected a 3-tier role-based access control (RBAC) system (Admin, Teacher, Student) using Supabase Row-Level Security(RLS), securing data isolation across **500+** monthly sessions.
- Reduced LLM API costs by **40%** and token usage by **35%** by executing a model migration and establishing an adaptive context window management strategy.
- Engineered a fault-tolerant speech-to-text pipeline utilizing Sarvam AI APIs and offline recording queues, achieving **100%** voice input uptime across **7** highly nuanced Indian dialects.

AI Research Intern

Jan 2024 – May 2024

PURPLESPOT, Remote

- Built an end-to-end NLP pipeline processing **5,000+** daily WhatsApp logistics messages into structured SQL database records, saving **15–20** operational hours/week.
- Boosted extraction accuracy to **90%+** by benchmarking **10+** LLMs and optimizing prompt design to handle code-switching, localized slang, and domain-specific abbreviations.

Machine Learning Intern

Jan 2023 – Dec 2023

DGTEL Rehumanice Pvt. Ltd., Remote

- Trained a custom CNN to **98%** accuracy on a proprietary **2,000+** sample Tamil handwritten character dataset using TensorFlow.
- Achieved **sub-50ms** offline inference on mobile devices by converting the trained model to TensorFlow Lite and optimizing for edge deployment with zero internet dependency.

PROJECTS

ScholarStream: Agentic RAG Pipeline | *Python, FastAPI, LangChain, LangGraph, OpenSearch, Redis, Docker* | [GitHub](#) 2026

- Architected and deployed a production-grade **7-service** backend system with ReAct-style agentic orchestration (decide → retrieve → grade → rewrite → generate), hybrid BM25+KNN search, and full Docker Compose deployment.
- Reduced query latency from **15–20s** to **100ms** (**150–400×** improvement) by engineering a Redis semantic caching layer with SHA-256 keyed lookups, eliminating redundant upstream LLM inference.
- Instrumented end-to-end observability using Langfuse v3 for per-request span tracing, tracking real-time token usage, execution latency, and hallucination metrics.

VeriTrace: AI Compliance Auditor | *Python, FastAPI, LangChain, Ollama, Llama 3, ChromaDB* | [GitHub](#) 2025

- Reduced compliance document lookup from **hours** to **seconds** by building an offline agentic Retrieval-Augmented Generation (RAG) system with semantic chunking and citation-backed responses.
- Enabled secure deployment in air-gapped environments by serving all inference on-premise with guardrails and hallucination mitigation, with zero data egress for enterprise workloads.

EDUCATION

University of California, Riverside

Riverside, CA

Master of Science in Robotics (Specialization in AI & Perception) | GPA: 3.97/4.0

Sept 2024 – Dec 2025

Relevant Coursework: Foundations of Machine Learning · Data Mining Techniques · Artificial Intelligence

CERTIFICATIONS

Claude Code in Action (Anthropic) | Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate